

Quant II

Lab 8

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Today's Agenda

- RDD
- Estimation and Inference
- Falsification Tests
- Local Randomization Approach
- Fuzzy RD
- Geographical RD

Setting

Sharp regression discontinuity design involves:

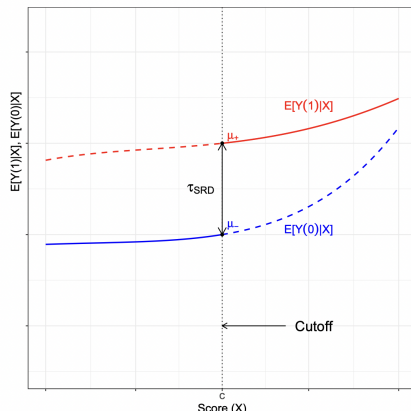
- Scalar continuous score (a.k.a. running variable, forcing variable) X_i
- Scalar cutoff c (with non-zero density of X_i on both sides)
- Binary treatment $D_i = 1[X_i \geq c]$
 - Fully determined by the score, no discretion
 - Rule is discontinuous in X_i at c
- Continuity of (expected) potential outcomes $\mathbb{E}[Y(0)|X_i], \mathbb{E}[Y(1)|X_i]$ at $X_i = c$
 - No other determinant of Y_i jumps at $X_i = c$
 - Score cannot be precisely (and endogenously) manipulated

Identification

Then

$$\lim_{X_i \downarrow c} \mathbb{E}[Y_i | X_i = x] - \lim_{X_i \uparrow c} \mathbb{E}[Y_i | X_i = x] = \mathbb{E}[Y_i(1) - Y_i(0) | X_i = c]$$

- Discontinuity of regression $\mathbb{E}[Y_i | X_i]$ at c identifies a local causal parameter



Is RDD a special case of something?

- Is RDD like selection on observables, with X_i as a control?
- Is RDD like IV, with X_i as instrument?
- Is RDD like a RCT in the neighborhood of $X_i = c$?

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- Is RDD like IV, with X_i as instrument?
 - No. Exogeneity of X_i is not assumed, e.g. higher vote share in election $t - 1$ correlates with higher vote share in t
- Is RDD like a RCT in the neighborhood of $X_i = c$?
 - Yes. Continuity of potential outcomes implies their balance around $X_i = c$
 - No. This only holds in an infinitesimal neighborhood. So we need to be careful with estimation

Econometrica, Vol. 82, No. 1 (January, 2014), 229–269

ISLAMIC RULE AND THE EMPOWERMENT OF THE POOR AND PIOUS

BY ERIK MEYERSSON¹

Does Islamic political control affect women's empowerment? Several countries have recently experienced Islamic parties coming to power through democratic elections. Due to strong support among religious conservatives, constituencies with Islamic rule often tend to exhibit poor women's rights. Whether this reflects a causal relationship or a spurious one has so far gone unexplored. I provide the first piece of evidence using a new and unique data set of Turkish municipalities. In 1994, an Islamic party won multiple municipal mayor seats across the country. Using a regression discontinuity (RD) design, I compare municipalities where this Islamic party barely won or lost elections. Despite negative raw correlations, the RD results reveal that, over a period of six years, Islamic rule increased female secular high school education. Corresponding effects for men are systematically smaller and less precise. In the longer run, the effect on female education remained persistent up to 17 years after, and also reduced adolescent marriages. An analysis of long-run political effects of Islamic rule shows increased female political participation and an overall decrease in Islamic political preferences. The results are consistent with an explanation that emphasizes the Islamic party's effectiveness in overcoming barriers to female entry for the poor and pious.

Meyersson (2014)

Causal effects of interest:

- Victory of Islamic candidate on educational attainment of women

Elements:

- **Outcome (Y)**: percentage of women aged 15-20 in 2000 who had completed high school by 2000
- **Running variable (X)**: vote percentage of the Islamic party minus vote percentage of the strongest secular opponent
- **Treatment (T)**: 1 if Islamic party won in 1994, 0 otherwise

A checklist for how to support the RDD analysis

- A graphical representation and test of “balance” and first stage (if fuzzy)
- Permutation test of characteristic at cutoff
- The density of the forcing variable
- Placebo checks
- A graphical representation of the outcomes (what we’ve already seen)
- Estimates based on optimal bandwidth choice and robust inference, using local linear analysis
 - These decisions vary depending on running variable. If discrete running variable, need to account for discreteness (Kolesar and Rothe (2018))
 - Should use local linear regression, and not global polynomials (Gelman and Imbens)
- Robustness analysis along bandwidth choice (and other tuning parameters)
 - Present this graphically

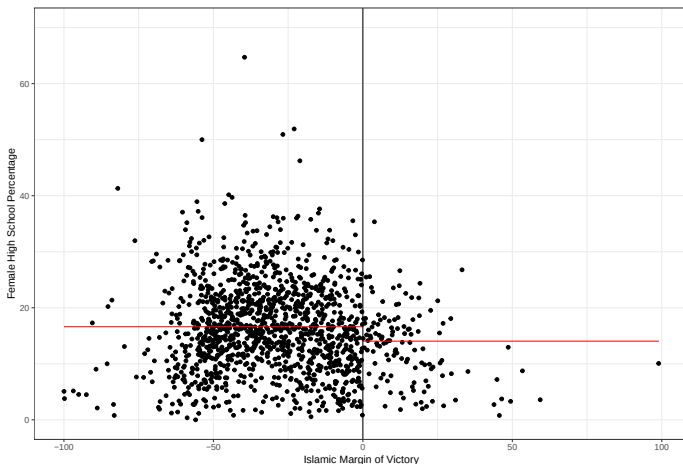
Set-up

```
library(rdrobust)
library(rddensity)
library(haven)
library(ggplot2)
# Import data and define variables
data <- read_dta("CIT_2019_Cambridge_polecon.dta")

Y <- data$Y
X <- data$X
t <- data$T
T_X <- t*X
```

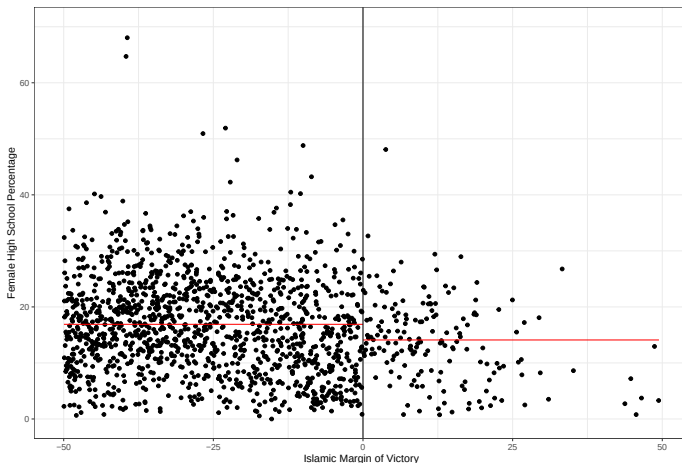
Visualization

```
# Raw means comparison  
rdplot(Y, X, nbins = c(2500, 500), p = 0, col.lines = "red", col.dots = "black", title = "",  
       x.label = "Islamic Margin of Victory", y.label = "Female High School Percentage", y.lim = c(0,70))
```



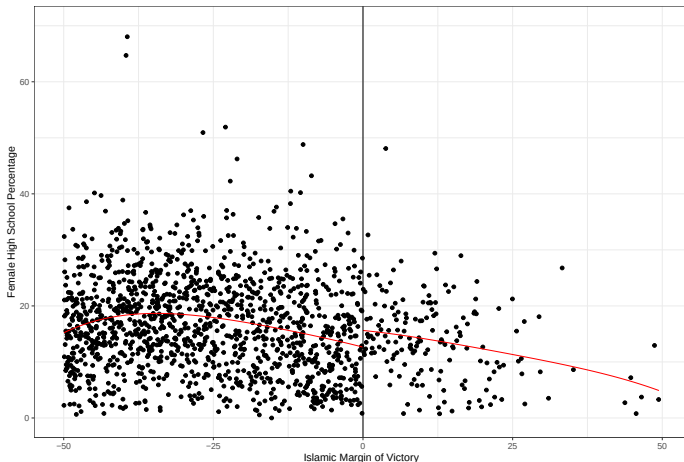
Visualization

```
# Local means comparison
rdplot(Y[abs(X) <= 50], X[abs(X) <= 50], nbins = c(2500, 500), p = 0, col.lines = "red", col.dots = "black",
       title = "", x.label = "Islamic Margin of Victory", y.label = "Female High School Percentage",
       y.lim = c(0,70))
```

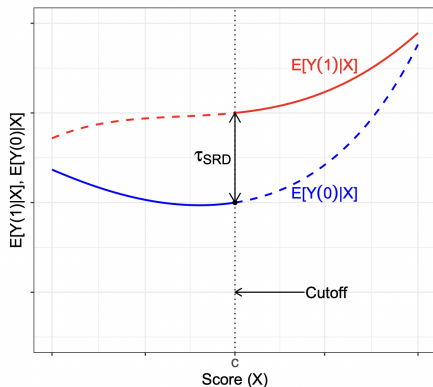


Visualization

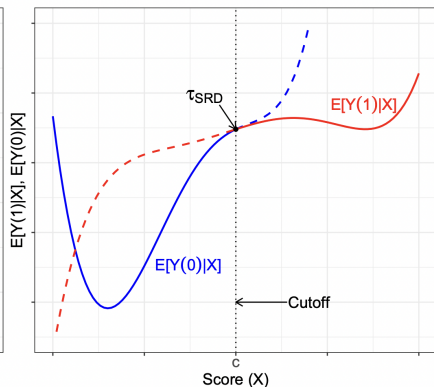
```
# Local means comparison
rdplot(Y[abs(X) <= 50], X[abs(X) <= 50], nbins = c(2500, 500), p = 4, col.lines = "red", col.dots = "black",
       title = "", x.label = "Islamic Margin of Victory", y.label = "Female High School Percentage",
       y.lim = c(0,70))
```



The Local Nature of RD Effects

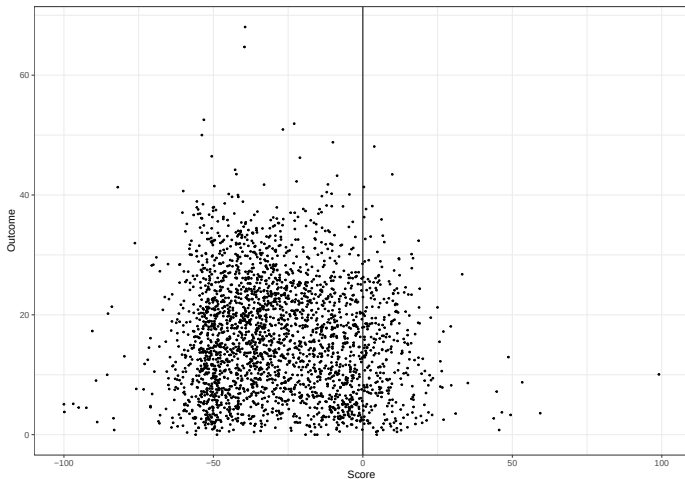


(a) Mild Heterogeneity



(b) Severe Heterogeneity

the Raw Data is Rarely Informative



RD plot: Details

Shows discontinuity in regression $\mathbb{E}[Y_i|X_i]$ in two ways:

- Smooth fit on each side: shape and nonlinearity of the regression
 - Separately on the left and right of c , fitted values from

$$Y_i = \alpha_0 + \alpha_1(X_i - c) + \dots + \alpha_p(X_i - c)^p + \text{error}$$

e.g. with quartic polynomial ($p = 4$)

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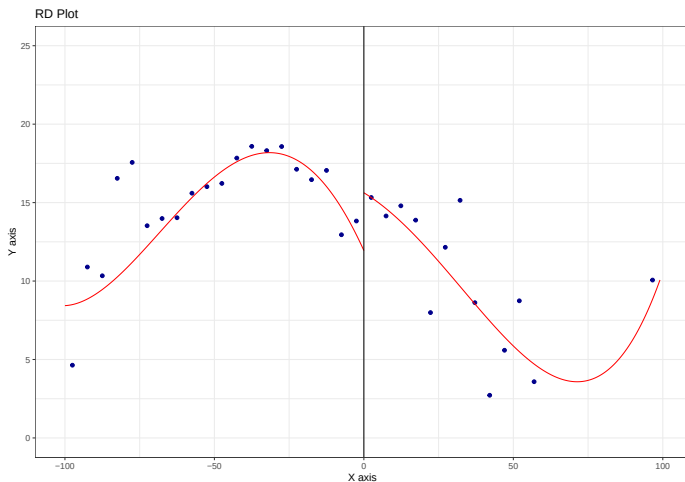
- Binscatter: a local/nonparametric estimator of the regression
 - Separately on each side, for some bins of X_i : average Y_i against bin's midpoint
 - Bins with similar numbers of observations (splitting by quantile) are more informative. But equal width is also common
 - Calonico, Cattaneo, Titiunik (2015) propose data-driven optimal number of bins

Partition of Islamic Margin of Victory into 40 Bins of Equal Length

Bin	Average Outcome in Bin	Number of Observations	Group Assignment
$\mathcal{B}_{-,1} = [-100, -95)$	$\bar{Y}_{-,1} = 4.6366$	4	Control
$\mathcal{B}_{-,2} = [-95, -90)$	$\bar{Y}_{-,2} = 10.8942$	2	Control
$\vdots$	$\vdots$	$\vdots$	$\vdots$
$\mathcal{B}_{-,19} = [-10, -5)$	$\bar{Y}_{-,19} = 12.9518$	149	Control
$\mathcal{B}_{-,20} = [-5, 0)$	$\bar{Y}_{-,20} = 13.8267$	148	Control
$\mathcal{B}_{+,1} = [0, 5)$	$\bar{Y}_{+,1} = 15.3678$	109	Treatment
$\mathcal{B}_{+,2} = [5, 10)$	$\bar{Y}_{+,2} = 13.9640$	83	Treatment
$\vdots$	$\vdots$	$\vdots$	$\vdots$
$\mathcal{B}_{+,19} = [90, 95)$	$\bar{Y}_{+,19} = \text{NA}$	0	Treatment
$\mathcal{B}_{+,20} = [95, 100]$	$\bar{Y}_{+,20} = 10.0629$	1	Treatment

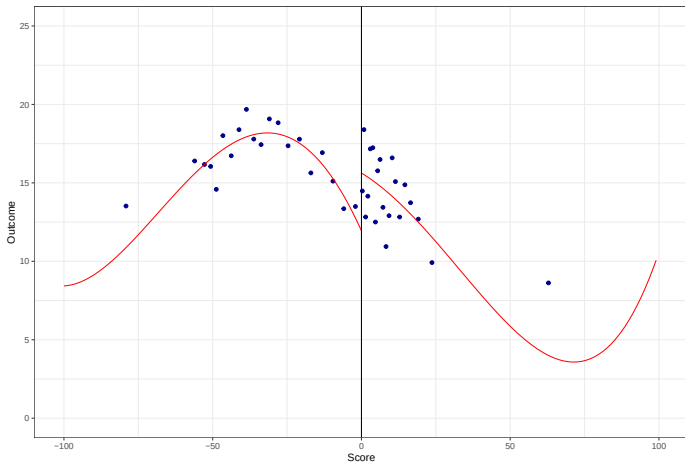
Evenly-Spaced Binning

```
# Default is 4th polynomial degree on each side  
rdplot(Y, X, nbins = c(20,20), binselect = "es", y.lim = c(0,25))
```



Quantile-Spaced Binning

```
rdplot(Y, X, nbins = c(20,20), binselect = 'qs', x.label = 'Score',  
       y.label = 'Outcome', title = '', x.lim = c(-100,100), y.lim = c(0,25))
```

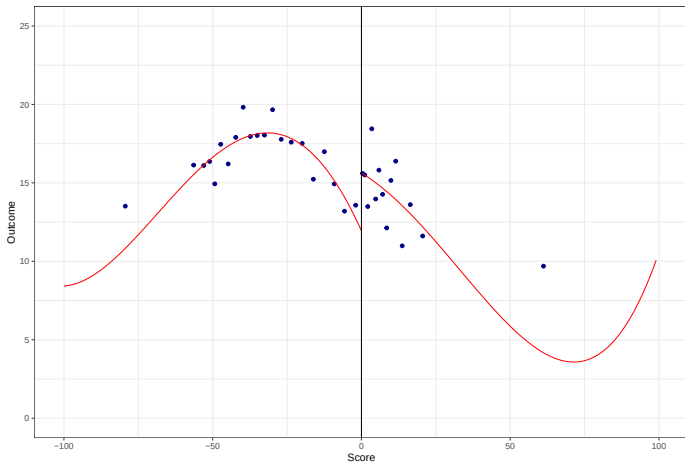


Choosing the Number of Bins

- Once we choose how to do bins, how should we choose the number?
 - Can we choose “optimally”?
- Cattaneo et al. (2020) argue for two approaches (available in `rdrobust`’s `rdplot`):
- IMSE-minimizing, and mimicking variance.
 - IMSE-minimizing trades off between bias and variance in choice of bins, but does it over the whole range – proportional to $n^{1/3}$
 - Mimicking-variance tries to match the underlying variance of the raw data in the binned plots – proportional to $\frac{n}{\log(n)^2}$

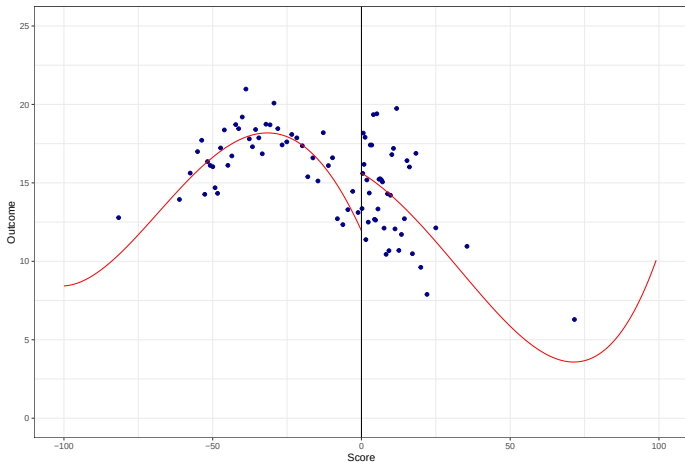
Integrated Mean Squared Error (IMSE) Method

```
rdplot(Y, X, binselect = 'qs', x.label = 'Score',  
       y.label = 'Outcome', title = '', x.lim = c(-100,100), y.lim = c(0,25))
```



Mimicking Variance Method

```
rdplot(Y, X, binselect = 'qsmv', x.label = 'Score',  
       y.label = 'Outcome', title = '', x.lim = c(-100,100), y.lim = c(0,25))
```

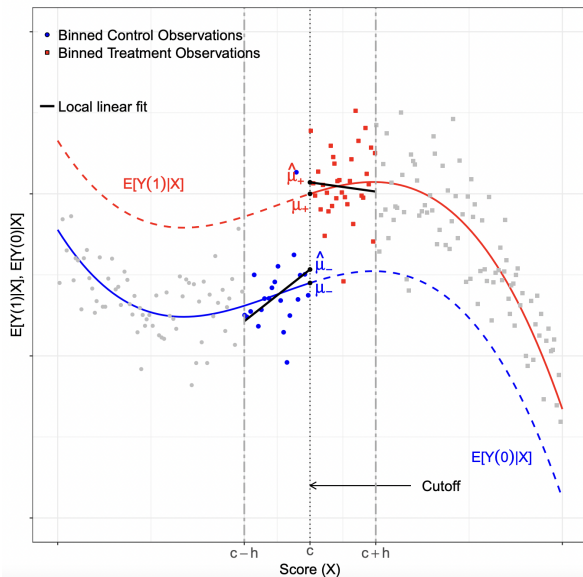


The Continuity-Based Approach to RD Analysis

Local Polynomial Approach:

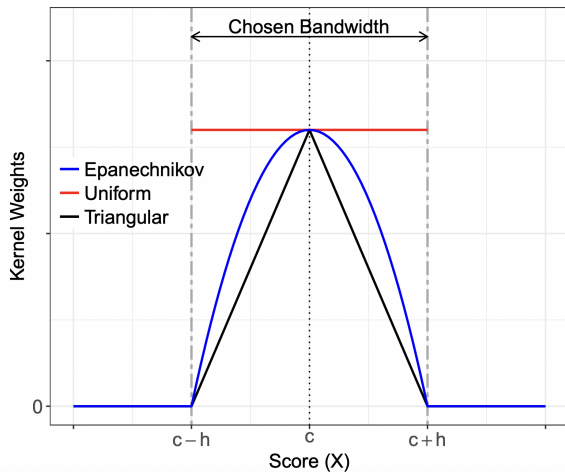
- Choose a polynomial order p and a kernel function $K(\cdot)$
- Choose a bandwidth h around the cutoff c
- On each side of the cutoff, fit a local polynomial by weighted least squares using observations within the bandwidth, with weights $K\left(\frac{X_i - c}{h}\right)$
- Compute the sharp RD estimate as the difference between the two fitted functions evaluated at the cutoff

RD Estimation with Local Polynomial



Kernel

Triangular one is recommended (weight = 0 outside h and $\uparrow$ as we get closer to c) and default in `rdrobust`. Alternatives: Uniform and Epanechnikov



Kernel

```
# By default c = 0
out <- rdrobust(Y, X, kernel = "uniform", p = 1)
summary(out)
```

Sharp RD estimates using local polynomial regression.

Number of Obs.	2629
BW type	mserd
Kernel	Uniform
VCE method	NN

Number of Obs.	2314	315
Eff. Number of Obs.	474	251
Order est. (p)	1	1
Order bias (q)	2	2
BW est. (h)	15.449	15.449
BW bias (b)	28.309	28.309
rho (h/b)	0.546	0.546
Unique Obs.	2311	315

	Point Estimate	Robust z	Inference P> z	[95% C.I.]
RD Effect	3.202	2.044	0.041	[0.134 , 6.339]

Kernel

```
# By default c = 0
out <- rdrobust(Y, X, kernel = "triangular", p = 1)
summary(out)
```

Sharp RD estimates using local polynomial regression.

Number of Obs.	2629	
BW type	mserd	
Kernel	Triangular	
VCE method	NN	
Number of Obs.	2314	315
Eff. Number of Obs.	529	266
Order est. (p)	1	1
Order bias (q)	2	2
BW est. (h)	17.240	17.240
BW bias (b)	28.576	28.576
rho (h/b)	0.603	0.603
Unique Obs.	2311	315

	Point Estimate	Robust z	Inference P> z	[95% C.I.]
RD Effect	3.020	1.776	0.076	[-0.309 , 6.276]

The choice of local polynomial order

- ① A polynomial of order zero has undesirable theoretical properties at boundary points, which is precisely where RD estimation must occur.
- ② For a given bandwidth, increasing the order of the polynomial generally improves the accuracy of the approximation but also increases the variability of the treatment effect estimator.
- ③ Higher-order polynomials tend to produce overfitting of the data and lead to unreliable results near boundary points.

The choice of local polynomial order

```
out <- rdrobust(Y, X, p = 1)
summary(out)
```

Sharp RD estimates using local polynomial regression.

Number of Obs.	2629	
BW type	mserd	
Kernel	Triangular	
VCE method	NN	
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Eff. Number of Obs.	529	266
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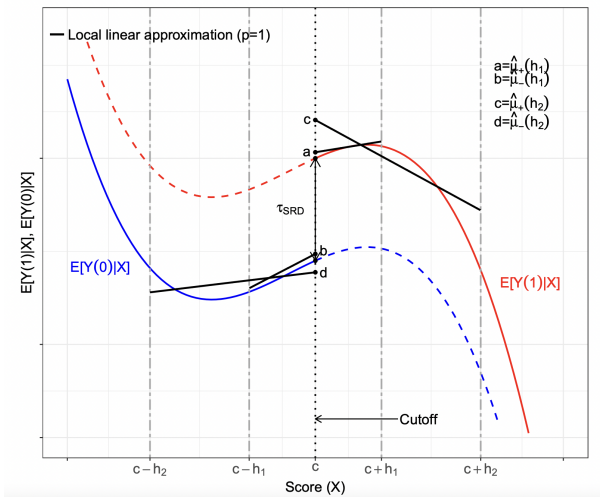
```
out <- rdrobust(Y, X, p = 2)
summary(out)
```

Sharp RD estimates using local polynomial regression.

Number of Obs.	2629	
BW type	mserd	
Kernel	Triangular	
VCE method	NN	
Number of Obs.	2314	315
Eff. Number of Obs.	702	291
Order est. (p)	2	2
Order bias (q)	3	3
BW est. (h)	23.121	23.121
BW bias (b)	35.191	35.191
rho (h/b)	0.657	0.657
Unique Obs.	2311	315

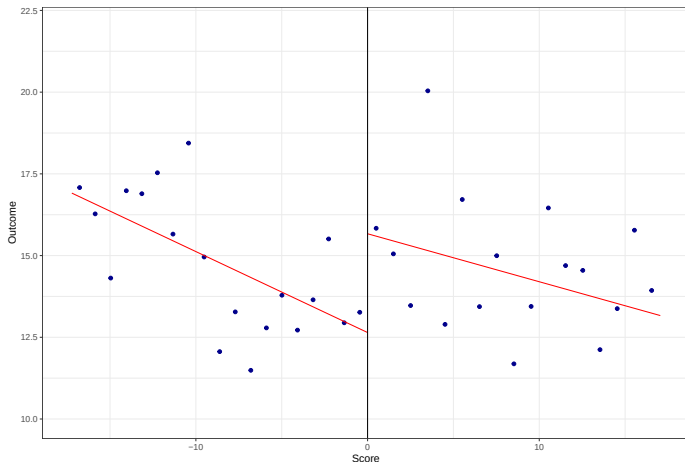
	Point Estimate	Robust z	Inference P> z	[95% C.I.]
RD Effect	2.772	1.325	0.185	[-1.276 , 6.600]

Bandwidth Selection



Optimal bandwidth choice

```
# mserd: MSE-optimal bandwidth selector for the RD treatment effect estimator.  
out <- rdrobust(Y, X, kernel = "triangular", p = 1, bwselect = "mserd")  
bw <- out$bws[1,1]  
rdplot(Y[abs(X)<=bw], X[abs(X)<=bw], p = 1, kernel = 'triangular',  
       x.label = 'Score', y.label = 'Outcome', title = '', y.lim = c(10,22))
```



Inference and bias correction

- Problem: h^* minimizes MSE by trading off bias² and variance
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- Problem: h^* minimizes MSE by trading off bias² and variance
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 - conventional confidence intervals typically have incorrect coverage.
- One solution: “undersmoothing”
 - Use bandwidth h much smaller than h^* .
 - Then inference is fine (bias $\ll$ SE)
 - But unclear how to choose h , and would be a higher-MSE estimator
- Better solution: **robust bias correction** \ (Calonico et al. 2014; rdrobust)
 - Fit the RD estimator using a low-order local polynomial
 - Estimate its leading bias using a higher-order local polynomial
 - Construct a confidence interval that accounts for both sampling variation and the noise in the bias estimate
 - “Robust bias-corrected conf. interval”

Inference

```
out <- rdrobust(Y, X, kernel = 'triangular', p = 1, bwselect = 'mserd', all = TRUE)
summary(out)
```

Sharp RD estimates using local polynomial regression.

Number of Obs.	2629	
BW type	mserd	
Kernel	Triangular	
VCE method	NN	
Number of Obs.	2314	315
Eff. Number of Obs.	529	266
Order est. (p)	1	1
Order bias (q)	2	2
BW est. (h)	17.240	17.240
BW bias (b)	28.576	28.576
rho (h/b)	0.603	0.603
Unique Obs.	2311	315

Method	Coef.	Std. Err.	z	P> z	[95% C.I.]
Conventional	3.020	1.427	2.116	0.034	[0.223 , 5.816]
Bias-Corrected	2.983	1.427	2.090	0.037	[0.186 , 5.780]
Robust	2.983	1.680	1.776	0.076	[-0.309 , 6.276]

RDD assumption violations

Sorting / manipulation of the running variable: units or relevant actors can precisely influence treatment assignment by manipulating the running variable around the cutoff.

- In our example: Islamic parties may invest more in winning election in some important municipalities, If they are successful, just-won towns would be different from just-lost ones, i.e. a discontinuity at the cutoff

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We can check for:

- ① the null treatment effect on predetermined covariates or placebo outcomes;
- ② the continuity of the score density around the cutoff;
- ③ the treatment effect at artificial cutoff values;
- ④ the exclusion of observations near the cutoff;
- ⑤ the sensitivity to bandwidth choices.

Checking for balance

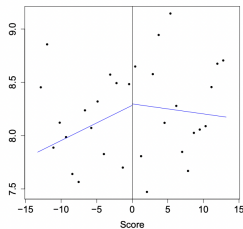
```
out = rdrobust(data$lpop1994, X)
summary(out)
```

Sharp RD estimates using local polynomial regression.

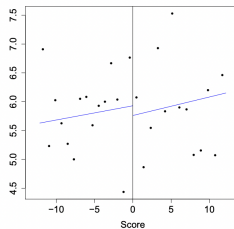
Number of Obs.	2629	
BW type	mserd	
Kernel	Triangular	
VCE method	NN	
Number of Obs.	2314	315
Eff. Number of Obs.	400	233
Order est. (p)	1	1
Order bias (q)	2	2
BW est. (h)	13.320	13.320
BW bias (b)	21.368	21.368
rho (h/b)	0.623	0.623
Unique Obs.	2311	315

	Point Estimate	Robust z	Inference P> z	[95% C.I.]
RD Effect	0.012	0.001	0.999	[-0.644 , 0.645]

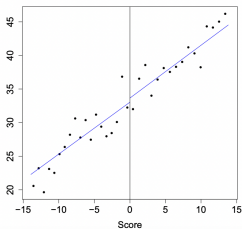
Graphical Illustration of Local Linear RD Effects for Predetermined Covariates



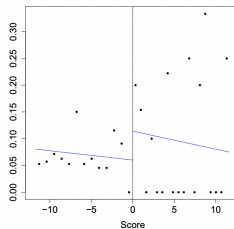
(a) Log Population in 1994



(b) Number of Parties Receiving Votes in 1994



(c) Islamic Vote Percentage in 1994



(d) Islamic Mayor in 1989

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- However
 - Testing just mean differences doesn't look at other parts of the distribution (which may more obviously violate this) and so may have low power
 - Because the local sample size is effectively small, this can create problematic inference issues if the function is not particularly smooth

Checking for balance

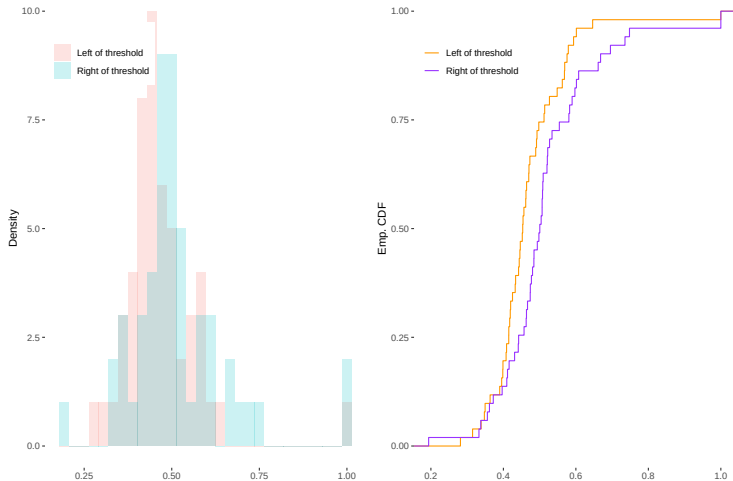
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- However
 - Testing just mean differences doesn't look at other parts of the distribution (which may more obviously violate this) and so may have low power
 - Because the local sample size is effectively small, this can create problematic inference issues if the function is not particularly smooth
- Canay and Kamat (2017) propose a permutation test
 - Key intuition – covariates should be *approximately* identically distributed on each side of the cutoff
 - This is an *asymptotic* argument, since it's not actually a random experiment

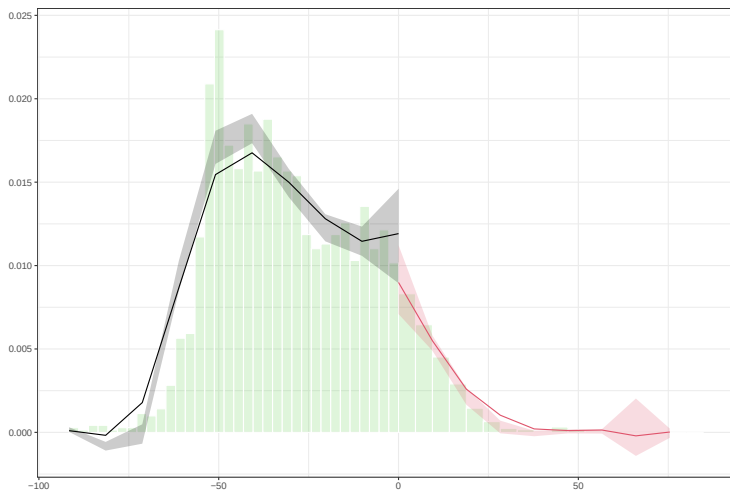
Checking for Balance - Canay and Kamat

```
library(RATest)
lee2008<-lee2008
set.seed(101)
permtest<-RDperm(W="demshareprev", z="difdemshare",data=lee2008,q_type=51)
plot(permtest,w="demshareprev")
```



Testing for sorting in running variable

```
library(rddensity)
denstest <- rddensity(data$X, c = 0)
rdplotdensity(denstest, data$X)
```



RDD with placebo cutoffs

```
placebo <- function(Y, X, new_cutoff){
  if (new_cutoff > 0){
    Y <- Y[X>=0]; X <- X[X>=0]
  }
  if (new_cutoff < 0){
    Y <- Y[X<0]; X <- X[X<0]
  }
  else{
    Y <- Y; X <- X
  }
  out <- rdrobust(Y, X, c = new_cutoff)
  coef <- out$coef["Conventional",]
  ll <- out$ci["Robust",1]
  ul <- out$ci["Robust",2]

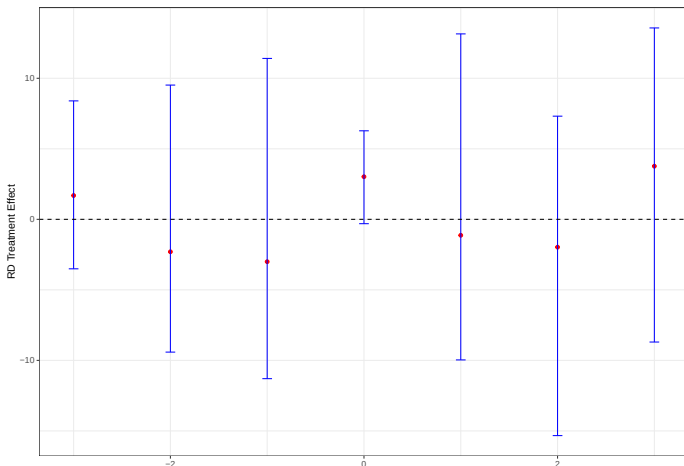
  cbind(coef, ll, ul)
}

cutoffs <- as.list(c(-3:3))
(placebos <- do.call("rbind", lapply(cutoffs, function(i) placebo(Y, X, i))))
```

	coef	ll	ul
[1,]	1.687817	-3.5083421	8.397096
[2,]	-2.300012	-9.4137423	9.517862
[3,]	-3.004159	-11.2961682	11.407925
[4,]	3.019526	-0.3092892	6.275769
[5,]	-1.130650	-9.9671538	13.146914
[6,]	-1.972790	-15.3333665	7.313371
[7,]	3.766433	-8.6998298	13.569346

RDD with placebo cutoffs

```
library(dplyr)
placebos %>% as.data.frame() %>% mutate(cutoff = -3:3) %>%
  ggplot(aes(x=cutoff, y=coef)) + geom_point(col="red") +
  geom_errorbar(aes(ymin=ll, ymax=ul), col="blue",width=0.1) +
  labs(y = "RD Treatment Effect", x = "Cutoff (x=0 true cutoff)") +
  geom_hline(yintercept=0, col="black", linetype = "dashed") + theme_bw()
```



Analogies with experiment

- Lee (2008): RD design can be as credible as a randomized experiment for units very near cutoff
- Imagine that score depends on each unit's unobservables characteristics and choices
- If the two following conditions hold:
 - some random variation in the score near the cutoff
 - probability of this random “error” doesn't change abruptly at cutoff
- Then the RD design can be seen as an experiment:
 - units barely above the cutoff as-if randomly assigned to treatment
 - units barely above the cutoff as-if randomly assigned to control
- This fails if individuals have ability to exactly control their score

Analogies with experiment

- Consider an RD Design where:
 - treatment is assigned based on score exceeding cutoff
 - units lack ability to manipulate score (continuity holds)
- **Crucial distinction:**
 - Experiment → no need to make assumptions about shape of the average potential outcomes
 - RD design → inferences depend crucially on assumptions regarding functional form of the conditional mean functions
- Any experiment can be recast as an RD design where
 - score is a uniform random variable
 - cutoff chosen to ensure a given probability of treatment
 - Ex: each student assigned uniform random number between 0 and 100, scholarship given to students whose score is above 50

Local Randomization Approach to RD Design

- Choose a narrow window around the cutoff and assume that, within this window, treatment assignment is **as-if random**
- Then units just above and just below the threshold can be compared like treated and control units in an experiment
- This is stronger than the continuity-based approach
 - it requires the running variable to matter little within the window apart from determining treatment status.

How to Implement It

Step 1: Select the window

- Start near the cutoff and choose the largest window that still passes balance checks on predetermined covariates.

How to Implement It

Step 1: Select the window

- Start near the cutoff and choose the largest window that still passes balance checks on predetermined covariates.

Step 2: Analyze as an experiment

- Within that window, compare treated and control units using randomization-based inference.
- Inference depends on the assumed assignment mechanism and is most credible when the window is small and balance is good.

Empirical Illustration: Incumbency Advantage (CFT, 2015, JCI)

- Problem: incumbency advantage (U.S. senate).
- Data:
 - Y_i = election outcome at $t + 1$.
 - T_i = whether party wins election at t .
 - X_i = margin of victory at t ($c = 0$).
 - Z_i = covariates (*demvoteshlag1*, *demvoteshlag2*, *dopen*, etc.).
- Potential outcomes:
 - $Y_i(0)$ = election outcome at $t + 1$ if had not been incumbent.
 - $Y_i(1)$ = election outcome at $t + 1$ if had been incumbent.

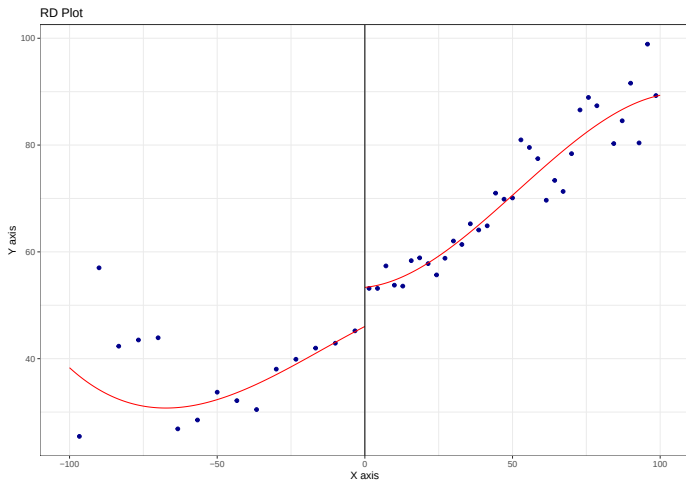
Empirical Illustration: Incumbency Advantage (CFT, 2015, JCI)

```
library(rdlocrand)
library(rdrobust)
data <- read.csv("rdlocrand_senate.csv")
X <- cbind(data$presdemvoteslag1,
           data$population/1000000, data$demvoteslag1,
           data$demvoteslag2, data$demwinprv1,
           data$demwinprv2, data$dopen,
           data$dmidterm, data$dpresdem)
colnames(X) <- c("DemPres Vote", "Population",
                "DemSen Vote t-1", "DemSen Vote t-2",
                "DemSen Win t-1", "DemSen Win t-2",
                "Open", "Midterm", "DemPres")

R <- data$demmv
Y <- data$demvotesfor2
D <- as.numeric(R>=0)
```

Simple RD plot

```
rdplot(Y,R,p=3)
```



Randomization inference using selected window

```
out <- rdrandinf(Y, R, wl = -2.5, wr = 2.5, seed = 50)
```

Selected window = [-2.5;2.5]

Running randomization-based test...

Randomization-based test complete.

```
Number of obs      =          1297
Order of poly      =              0
Kernel type       =      uniform
Reps              =          1000
Window            =      set by user
H0:               tau =          0.000
Randomization     = fixed margins
```

```
Cutoff c =      0.000   Left of c   Right of c
      Number of obs          595       702
      Eff. number of obs          63       57
      Mean of outcome        44.068      53.235
      S.d. of outcome        10.627      8.289
      Window                -2.500      2.500
```

```
=====
                        Finite sample          Large sample
                        -----
Statistic              T          P>|T|          P>|T|          Power vs d = 5.313
=====
Diff. in means         9.167          0.000          0.000          0.866
=====
```

Binomial randomization mechanism

```
bern_prob <- numeric(length(R))
bern_prob[abs(R) > 2.5] <- NA
bern_prob[abs(R) <= 2.5] <- 1/2
out <- rdrandinf(Y, R, wl = -2.5, wr = 2.5, seed = 50, bernoul
```

Selected window = [-2.5;2.5]

Running randomization-based test...

Randomization-based test complete.

Number of obs	=	120
Order of poly	=	0
Kernel type	=	uniform
Reps	=	1000

Binomial randomization mechanism

Selected window = [-2.5;2.5]

Running randomization-based test...

Randomization-based test complete.

Number of obs = 120
Order of poly = 0
Kernel type = uniform
Reps = 1000
Window = set by user
H0: tau = 0.000
Randomization = Bernoulli

Cutoff c = 0.000	Left of c	Right of c
Number of obs	63	57
Eff. number of obs	63	57
Mean of outcome	44.068	53.235
S.d. of outcome	10.627	8.289
Window	-2.500	2.500

Statistic	Finite sample		Large sample		Power vs d = 5.313
	T	P> T	P> T		
Diff. in means	9.167	0.000	0.000		0.866

Selecting the Window

- `rdwinselect()` searches over candidate windows around the cutoff and tests whether observations just below and above the threshold look balanced.

```
out <- rdwinselect(R, X, seed = 50, reps = 1000, wobs = 2)
```

Mass points detected in running variable

You may use `wmasspoints` option for constructing windows at each

Window selection for RD under local randomization

Number of obs	=	1390
Order of poly	=	0
Kernel type	=	uniform
Reps	=	1000
Testing method	=	rdrandinf

Selecting the Window

Mass points detected in running variable

You may use `wmasspoints` option for constructing windows at each mass point

Window selection for RD under local randomization

```
Number of obs   =      1390
Order of poly   =          0
Kernel type     =    uniform
Reps            =      1000
Testing method  =    rdrandinf
Balance test    =    diffmeans
```

```
Cutoff c =    0.000   Left of c   Right of c
Number of obs      640       750
1st percentile      7         7
5th percentile     32        37
10th percentile     64        75
20th percentile    127       149
```

Window		p-value	Var. name	Bin.test	Obs<c	Obs>=c
-0.4843	0.4843	0.259	DemSen.Vote.t.2	0.230	9	16
-0.5630	0.5630	0.274	DemSen.Vote.t.2	0.265	11	18
-0.6378	0.6378	0.446	Open	0.296	13	20
-0.7305	0.7305	0.322	DemSen.Win.t.1	0.256	15	23
-0.8485	0.8485	0.130	Open	0.211	16	25
-1.0751	1.0751	0.042	Open	0.111	18	30
-1.1684	1.1684	0.102	Open	0.169	21	32
-1.2699	1.2699	0.133	Open	0.185	23	34
-1.3068	1.3068	0.144	Open	0.245	25	35
-1.3758	1.3758	0.149	Open	0.464	30	37

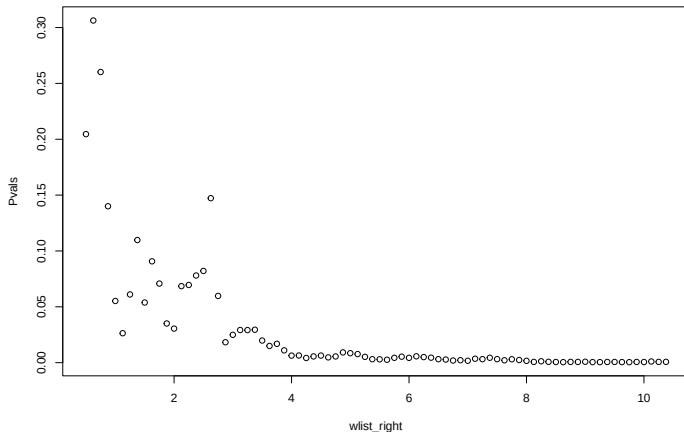
Recommended window is [-0.7305;0.7305] with 38 observations (15 below, 23 above).

P-value for Different Windows

```
tmp <- rdwinselect(R,X,wmin=.5,wstep=.125,approx=TRUE,nwin=80,quietly=TRUE,plot=TRUE)
```

Mass points detected in running variable

You may use `wmasspoints` option for constructing windows at each mass point



Fuzzy RD

- In a sharp RD, crossing the cutoff changes treatment status from 0 to 1 exactly.
- In a fuzzy RD, crossing the cutoff only changes the **probability** of treatment:
 - some units above the cutoff do not take treatment
 - some units below the cutoff still receive treatment
 - **no defiers**
- This makes fuzzy RD a local IV
 - the cutoff affects treatment take-up
 - treatment take-up affects the outcome
 - so the cutoff can be used as an instrument for actual treatment near the threshold

Fuzzy RD

- Interpretation:
 - first estimate the jump in the outcome at the cutoff
 - then scale it by the jump in treatment take-up
 - this gives the treatment effect for **compliers at the cutoff**
- `rdrobust(..., fuzzy = , ...)`

Geographic Regression Discontinuity

GRD geographic or administrative boundary splits units into treated and control areas and analysts make the case that the division into treated and control areas occurs in an as-if random fashion.

Peculiarities of GRD (Keele and Titiunik, 2015)

- ① *Compound* treatments
 - multiple treatments that affect the outcome of interest simultaneously
- ② Different measures of distance from the cutoffs may require different identification assumptions and affect fundamentally the interpretation of the results
- ③ Spatial variation in treatment effects can be mapped to specific locations, which can be used to detect geographic areas where the identification assumptions are more (or less) likely to hold

Recommendations for practice

- **Data:**

- researchers need to collect geographic data (addresses, latitude and longitude, or other geographic information that can be used for geocoding) along with more traditional covariates;
- qualitative research on the history of the border and conditions around it will often prove useful to justify various assumptions.

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- researchers should at least rule out nonzero treatment effects on predetermined covariates, which can be easily implemented using covariates as outcomes in the estimation for each boundary point.

Recommendations for practice

- **Data:**

- researchers need to collect geographic data (addresses, latitude and longitude, or other geographic information that can be used for geocoding) along with more traditional covariates;
- qualitative research on the history of the border and conditions around it will often prove useful to justify various assumptions.

- **Falsification tests:**

- researchers should at least rule out nonzero treatment effects on predetermined covariates, which can be easily implemented using covariates as outcomes in the estimation for each boundary point.

- **Isolating the treatment:**

- restrict the analysis to areas around the border where other important geographically defined institutional units are kept constant on either side of border.

rd2d Package Demo

[rd2d-demo]